

# Algorithmic Pricing: Carvana and Zillow

## Week 2

### The Algorithmic Pricing Problem

Pricing a used car or a home is not like pricing a commodity. Each asset has a distinct set of characteristics that affect its value, and those characteristics interact in ways that are hard to quantify. A used car's price might depend on make, model, mileage, condition, trim level, and local market demand. A home's price depends on square footage, neighborhood, school district, condition, and dozens of other factors. Traditional pricing relies on human judgment and comparable sales, which can lead to errors and is difficult to scale.

Carvana and Zillow both built machine learning models to price these assets at scale. Carvana, an online used car dealer, uses its models to generate instant purchase offers based on a vehicle's specifications, condition, mileage, and local market conditions. A seller receives an offer within minutes. The business model depends on pricing accurately enough to buy cars and resell them at a profit, eliminating the need for physical dealerships.

Zillow applied the same logic to real estate. Already known for its Zestimate tool, which estimates home values for over 100 million properties, Zillow launched Zillow Offers in 2018 to act on those estimates by purchasing homes directly. The plan was to buy properties, make light renovations, and resell quickly for a profit. By late 2021, the program was acquiring thousands of homes per month across 25 metropolitan areas.

Both companies were making the same fundamental bet: that models trained on historical transaction data could price heterogeneous assets accurately enough to buy and sell at scale, replacing human intermediaries with algorithms.

## Two Bets, Two Outcomes

Carvana's pricing models were tested severely during the pandemic. New car production stalled due to semiconductor shortages, driving demand into the used car market. Wholesale used vehicle prices rose more than 50% from early 2020 to their peak in early 2022, according to the Manheim Used Vehicle Value Index. When the market corrected, Carvana found itself holding inventory purchased at elevated prices. The company's stock fell 98% from its peak in August 2021, and it came close to insolvency but survived.

Zillow was not as fortunate. In the third quarter of 2021, the company ended with 9,790 homes in inventory but sold only 3,032, an average loss of roughly \$80,000 per home. In November 2021, Zillow shut down the program entirely, wrote off more than \$540 million, and laid off roughly 2,000 employees — 25% of its workforce.

## Discussion Questions

**Question 1.** Carvana's business model depends on its ability to generate accurate vehicle price predictions at scale. What strategic advantages does this capability provide compared to traditional dealerships?

**Question 2.** Both companies used machine learning to price assets, but Carvana's business model survived while Zillow Offers did not. What might explain the difference in outcomes?

## Sources

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